Offer #2026-09717

Research Internship / Priming Preference Elicitation with Goal-Conditioned Reinforcement Learning (F/M)

Contract type : Internship agreement

Level of qualifications required : Master's or equivalent

Fonction : Internship Research

About the research centre or Inria department

The Inria University of Lille centre, created in 2008, employs 360 people including 305 scientists in 15 research teams. Recognised for its strong involvement in the socio-economic development of the Hauts-De-France region, the Inria University of Lille centre pursues a close relationship with large companies and SMEs. By promoting synergies between researchers and industrialists, Inria participates in the transfer of skills and expertise in digital technologies and provides access to the best European and international research for the benefit of innovation and companies, particularly in the region.

For more than 10 years, the Inria University of Lille centre has been located at the heart of Lille's university and scientific ecosystem, as well as at the heart of Frenchtech, with a technology showroom based on Avenue de Bretagne in Lille, on the EuraTechnologies site of economic excellence dedicated to information and communication technologies (ICT)

Assignment

At Scool, Gautron (2022) have turned a high-fidelity crop simulator into an RL environment. In this problem, an AI advises a farmer throughout a harvesting season, deciding daily how much should the farmer water, fertilize, and so on, with a goal of striking a balance between several criteria such as yield or nitrate pollution under varying weather conditions. By running an off-the-shelf deep RL algorithm such as PPO (Schulman et al., 2017), it was shown in Gautron (2022) that RL can find more efficient solutions than human expert policies. However, the main drawback of the current decision support system is that it provides recommendations under a pre-defined trade-off between the different criteria (such as yield, pollution or work load) and can thus not adapt to the varying needs of individual farmers. An existing solution in the literature is to wrap an RL solver around a preference elicitation mechanism to allow non-RL-expert users to tune the reward function to their needs, while only interacting with the AI at a very abstract level. This is the so called Preference-based RL (PbRL, Wirth et al. (2017)), also known as RL from human feedback (RLHF). These methods have had a recent surge of popularity as they were shown to be useful for training large language models (Ouyang et al., 2022). An illustration of the RL from Human Feedback (RLHF) framework is given in Figure 1 and a survey can be found in Wirth et al. (2017).

Despite continuous efforts to improve PbRL algorithms (Hu et al., 2024; Zhu et al., 2025; Driss et al.), in their current state, they remain inadequate for real-world applications such as the aforementioned task. One of the limitation of current PbRL methods is that they use a costly policy optimization step using deep RL between each query round making the overall interaction last potentially several hours. The idea of this M2 internship is to exploit specificities of the task, namely that preferences can be expressed as proximity to a goal (a vector containing a target average crop yield, amount of used fertilizer, etc.), and use an unsupervised training phase with goal-conditioned RL (Liu et al., 2022) to learn quantities and models that can speed-up PbRL. These models include for instance a prior about possible goals, a set of pre-computed queries and a goal-conditioned policy able to reach target goals and adapt faster to specific user preferences.

For more information, please see Scool's job offer website https://team.inria.fr/scool/job-offers/

Main activities

Main Acitivies:

- Perform a literature review of PbRL and goal-conditioned RL and propose a model of users and user responses in PbRL from a goal-conditioned RL perspective.
- Use existing goal-conditioned RL algorithms on the gym-dssat task to learn goal-conditioned policies and a prior over achievable goals.
- Propose an approach for using the goal-conditioned policy and the goal prior to speed-up query generation and policy optimization in PbRL. Compare against existing PbRL baselines and evaluate improvements in compute time and reduction in the number of queries.

Optional Activities:

- Review the literature of preference elicitation with a focus on Bayesian aproaches such as (Viappiani and Boutilier, 2010).
- With the previously learned prior and user response model, develop a Bayesian query generation mechanism and evaluate its performance against existing query selection mechanisms in PbRL.

Benefits package

- Subsidized meals
- Partial reimbursement of public transport costs
- Leave (on a full time annual basis): 7 weeks of annual leave

- Possibility of teleworking and flexible organization of working hours
- Professional equipment available (videoconferencing, loan of computer equipment, etc.)
- Social, cultural and sports events and activities
- Access to vocational training
- Social security coverage

Remuneration

In accordance with current regulations

General Information

- **Theme/Domain :** Optimization, machine learning and statistical methods Statistics (Big data) (BAP E)
- **Town/city :** Villeneuve d'Ascq
- **Inria Center :** Centre Inria de l'Université de Lille
- **Starting date :** 2026-04-01
- **Duration of contract :** 6 months
- **Deadline to apply :** 2026-04-25

Contacts

- **Inria Team :** SCOOL
- **Recruiter :**
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About Inria

Inria is the French national research institute dedicated to digital science and technology. It employs 2,600 people. Its 200 agile project teams, generally run jointly with academic partners, include more than 3,500 scientists and engineers working to meet the challenges of digital technology, often at the interface with other disciplines. The Institute also employs numerous talents in over forty different professions. 900 research support staff contribute to the preparation and development of scientific and entrepreneurial projects that have a worldwide impact.

Warning : you must enter your e-mail address in order to save your application to Inria. Applications must be submitted online on the Inria website. Processing of applications sent from other channels is not guaranteed.

Instruction to apply

Please send your CV and cover letter

Defence Security :
This position is likely to be situated in a restricted area (ZRR), as defined in Decree

No. 2011-1425 relating to the protection of national scientific and technical potential (PPST).Authorisation to enter an area is granted by the director of the unit, following a favourable Ministerial decision, as defined in the decree of 3 July 2012 relating to the PPST. An unfavourable Ministerial decision in respect of a position situated in a ZRR would result in the cancellation of the appointment.

Recruitment Policy :
As part of its diversity policy, all Inria positions are accessible to people with disabilities.